

Exercise 2.4 Solution

Problem

If the step-size parameters, α_n , are not constant, then the estimate Q_n is a weighted average of previously received rewards with a weighting different from that given by equation (2.6). What is the weighting on each prior reward for the general case, analogous to (2.6), in terms of the sequence of step-size parameters?

Solution

The general incremental update is

$$Q_{n+1} = Q_n + \alpha_n(R_n - Q_n). \quad (1)$$

Equivalently,

$$Q_{n+1} = (1 - \alpha_n)Q_n + \alpha_n R_n. \quad (2)$$

Now unroll the recursion:

$$\begin{aligned} Q_{n+1} &= \alpha_n R_n + (1 - \alpha_n)Q_n \\ &= \alpha_n R_n + (1 - \alpha_n)[\alpha_{n-1}R_{n-1} + (1 - \alpha_{n-1})Q_{n-1}] \\ &= \alpha_n R_n + (1 - \alpha_n)\alpha_{n-1}R_{n-1} + (1 - \alpha_n)(1 - \alpha_{n-1})Q_{n-1} \\ &= \dots \end{aligned}$$

Continuing this expansion gives

$$Q_{n+1} = \left(\prod_{j=1}^n (1 - \alpha_j) \right) Q_1 + \sum_{i=1}^n \left[\alpha_i \prod_{j=i+1}^n (1 - \alpha_j) \right] R_i \quad (3)$$

where an empty product is defined as

$$\prod_{j=n+1}^n (1 - \alpha_j) = 1. \quad (4)$$

Therefore, the weight on reward R_i in Q_{n+1} is

$$\alpha_i \prod_{j=i+1}^n (1 - \alpha_j) \quad (5)$$

and the weight on the initial estimate Q_1 is

$$\prod_{j=1}^n (1 - \alpha_j). \quad (6)$$

Equivalently, since Q_n is based on rewards $R_1, \dots, R_{n-1}$, we can write

$$Q_n = \left(\prod_{j=1}^{n-1} (1 - \alpha_j) \right) Q_1 + \sum_{i=1}^{n-1} \left[\alpha_i \prod_{j=i+1}^{n-1} (1 - \alpha_j) \right] R_i \quad (7)$$

Thus, the weight on prior reward R_i in Q_n is

$$\alpha_i \prod_{j=i+1}^{n-1} (1 - \alpha_j). \quad (8)$$

Connection to the Constant Step-Size Case

If all step-size parameters are constant, so that $\alpha_i = \alpha$ for every i , then the general weight becomes

$$\alpha_i \prod_{j=i+1}^n (1 - \alpha_j) = \alpha(1 - \alpha)^{n-i}. \quad (9)$$

This recovers equation (2.6):

$$Q_{n+1} = (1 - \alpha)^n Q_1 + \sum_{i=1}^n \alpha(1 - \alpha)^{n-i} R_i. \quad (10)$$